

- Q.32 Write about manufacturing of ammonium nitrate.
 Q.33 Explain manufacturing of beer and wine (process only).
 Q.34 Describe storage vessels and their uses.
 Q.35 Explain working of heat exchanger with diagram.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain manufacturing of urea with flow sheet, uses and engineering problems.
 Q.37 Describe manufacturing of sugar with flow sheet and problems in production.
 Q.38 Explain manufacturing of Portland cement with flow sheet and uses.

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4th Sem / Chemical Engineering

Sub : Chemical Technology

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 P & ID stands for :
 a) Process Instrument Diagram
 b) Piping and Instrumentation Diagram
 c) Process Industry Design
 d) None
- Q.2 Sweetest sugar is:
 a) Glucose b) Fructose
 c) Sucrose d) Maltose
- Q.3 Kraft process is related to:
 a) Fertilizer b) Paper Industry
 c) Cement industry d) Sugar industry
- Q.4 Ethanol is produced by:
 a) Wet process b) Fermentation
 c) Hydrogenation d) Carbonation
- Q.5 Ethanol is produced by:
 a) Wet process b) Fermentation
 c) Hydrogenation d) Carbonation

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- Q.6 Raw material for sulphuric acid is:
- Limestone
 - Sulphur
 - Gypsum
 - Molasses
- Q.7 Urea is produced from:
- $\text{NH}_3 + \text{CO}_2$
 - $\text{H}_2\text{SO}_4 + \text{NH}_3$
 - $\text{CaCO}_3 + \text{H}_2\text{O}$
 - $\text{NaOH} + \text{Cl}_2$
- Q.8 Cement is manufactured mainly by:
- Fermentation
 - Dry process
 - Extraction
 - Distillation
- Q.9 Bleaching agents used in pulp industry:
- Chlorine
 - Hydrogen peroxide
 - Hypochlorite
 - All of these
- Q.10 Steam is generated in :
- Compressor
 - Boiler
 - Reactor
 - Separator

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define reactor.
- Q.12 Define detergent.
- Q.13 Name one fertilizer containing phosphorus.
- Q.14 Write one use of urea.
- Q.15 What is range of alkali?
- Q.16 What is glycerine?

- Q.17 Write formula of ammonia.
- Q.18 What is bagasse?
- Q.19 Define fermentation.
- Q.20 Name one type of flow sheet.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain different types of flow sheets used in industry.
- Q.22 Describe continuous hydrolysis and saponification process.
- Q.23 Explain manufacture of kraft paper.
- Q.24 Differentiate mechanical pulping and chemical pulping.
- Q.25 Discuss manufacturing steps of sugar from sugarcane.
- Q.26 Explain manufacturing of triple super phosphate with flow sheet.
- Q.27 Write about NPK fertilizers and their importance.
- Q.28 Discuss major engineering problems in cement industry.
- Q.29 Explain working principle of mixers.
- Q.30 Explain dry process of Portland cement.
- Q.31 Discuss importance of sulphuric acid in fertilizer industry.